

CSE2430 - Operating Systems

Homework IO and Storage (8)

Assigned on: **Mar 25 2022**

1 Heap Memory Management

Consider a memory partition of 100 kB, 400 kB, 200 kB, and 500 kB blocks (in this order), how would the following sequence of memory requests:

200 kB, 380 kB, 100 kB, 10 kB

be served using

- a) First fit
- b) Best fit
- c) Worst fit.

2 Hard Disk Capacity

A hard disk drive has sectors of 512 bytes, 8 platters, and 63 sectors per track. How many tracks does the disk have per surface if the capacity of the disk is 320 GB?

3 Disk Scheduling

The disk head of a disk with 100 tracks is initially at position 50 and the following sequence of requests comes in:

92, 23, 80, 17, 44, 60, 99, 71, 11

Show the schedule for the

- a) SSTF algorithm
- b) SCAN algorithm (assuming that the head at start is moving towards the innermost sector)

Turn-in Instructions

No hand-in required. Questions are meant for self-directed exam preparation and can be discussed during office hours and in the Q&A session at the end of the course.